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Natural Language Processing (AFI507)

A Comparative Research Report on Five Recent Studies

Academic Year: 2025–2026
Department of Computer Science & Engineering

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26/AFI/024

ADVANCES IN NATURAL LANGUAGE PROCESSING

A Comparative Research Report on Five Recent Studies

(2024-2026)

Abstract

Natural Language Processing (NLP) has recently transitioned from task-specific feature engineering and recurrent deep learning architectures to massive pre-trained Transformers and Large Language Models (LLMs). While modern generative foundation models show outstanding qualitative capabilities across general benchmarks, their direct application to specialized sequence labeling, high-throughput text classification, low-resource non-Latin morphologies, domain-specific scientific embeddings, and large-scale industrial aspect extraction exposes critical systemic challenges. This comparative survey synthesizes five foundational research papers published between 2024 and 2026: (1) a 42-page comprehensive Named Entity Recognition (NER) survey and multi-corpus empirical study (Keraghel et al., 2024); (2) an extensive 32-dataset, 8-language benchmarking of text classification paradigms in the LLM era (Vajjala & Shimangaud, 2025); (3) an empirical analysis of subword tokenization algorithms for the low-resource Himalayan language Dzongkha (Hall & Gonsalves, 2025); (4) an intrinsic and extrinsic evaluation suite comparing static domain embeddings with Transformers across scientific and biomedical tasks (Ahmed et al., 2025); and (5) a scalable neuro-symbolic framework coupling LLM zero-shot aspect discovery with fast, cost-effective classical classifiers applied across 4.7 million Yelp reviews (Patil et al., 2026). Across these diverse inquiries, a consistent finding emerges: monolithic, general-purpose LLM prompting is often neither the most accurate nor the most cost-effective solution. Instead, domain-adapted fine-tuning, specialized subword vocabularies, few-shot contrastive learning, and hybrid LLM-classical architectures demonstrate superior practical reliability and parameter efficiency.

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1. Introduction

The landscape of Natural Language Processing (NLP) is characterized by a persistent tension between general-purpose representation learning and specialized, task-specific performance constraints. The field's progression over the past decade can be broadly categorized into three distinct technical epochs: (i) the statistical and feature-engineering era, where models such as Hidden Markov Models (HMM), Conditional Random Fields (CRF), and Support Vector Machines (SVM) depended heavily on hand-crafted lexicons, orthographic heuristics, and domain gazetteers; (ii) the neural word representation and deep sequence modeling era, sparked by distributed static embeddings (Word2Vec, GloVe, FastText) integrated with Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory networks (BiLSTM-CRF); and (iii) the pre-trained Transformer and Large Language Model (LLM) revolution, initiated by masked auto-encoders (BERT, RoBERTa, DeBERTa) and culminating in massive autoregressive decoders (GPT-4, Llama-3, Qwen-2.5, Aya).

Although commercial generative models demonstrate remarkable fluent text generation and broad semantic generalization, their dominance is far from universal. In specialized sequence labeling, multi-class categorical classification across diverse scripts, low-resource subword segmentation, and cost-constrained production environments, large decoder-only models frequently suffer from excessive latency, tokenization bottlenecks, prompt sensitivity, boundary drift, and high monetary overhead. This report provides an in-depth comparative review of five research papers published between 2024 and 2026 that rigorously address these frontiers across diverse datasets, languages, and technical modalities.

2. Objectives of the Report

This survey report fulfills four primary academic and engineering objectives:

- Identify the foundational motivations, mathematical methodologies, and research gaps addressed by each of the five selected papers.
- Perform a rigorous breakdown of the model architectures, feature vectorization techniques, fine-tuning protocols, and experimental procedures reported in each investigation.
- Document and analyze the datasets, evaluation metrics (e.g., Macro F1, subword fertility, normalized sequence length, Pearson/Spearman correlation, regression R-squared), and key quantitative findings.
- Synthesize the comparative performance trends across all studies to establish concrete architectural guidelines regarding parameter efficiency, low-resource linguistic adaptation, domain specificity, and industrial scalability.

3. Cross-Cutting Motivations and Research Gaps

Despite addressing distinct NLP subfields, the five studies share profound underlying motivations stemming from the operational limitations of the prevailing LLM paradigm:

Core Challenge / Gap	Technical Limitation in Prior Art	Manifestation Across Surveyed Papers
General LLM Inefficiency in Sequence & Structured Tasks	Autoregressive LLMs generate text token-by-token and lack explicit probabilistic transition constraints (e.g., CRF layers), making token-level entity boundary detection imprecise.	Paper 1 proves GPT-4o falls substantially behind fine-tuned DeBERTa-v3 and GLiNER on NER. Paper 4 shows general Transformers trail SciBERT on scientific NER.
The Low-Resource Linguistic Divide & Script Bias	Multilingual LLMs and pre-trained subword tokenizers are trained overwhelmingly on high-resource Latin text, causing severe subword over-segmentation on non-Latin scripts.	Paper 3 documents massive subword fertility in Dzongkha under LLaMA/GPT-4o tokenizers. Paper 2 exposes 11%–29% performance penalties on Arabic and Hindi.
Classification Cardinality & Prompt Complexity Collapse	In-context prompting struggles as label sets grow (e.g., >15 classes), causing confusion and output format violations.	Paper 2 demonstrates that while zero-shot LLMs match baselines on 3-class sentiment, they lag full-data classifiers by >15% on 60-class intent detection.
Prohibitive Production Latency & Inference Budgets	Prompting billion-parameter LLMs across millions of production documents incurs unsustainable API billing, compute footprints, and high latency.	Paper 5 designs a hybrid LLM-classical pipeline to process 4.7M Yelp reviews. Paper 2 shows FastFit trains in minutes locally on a laptop.
Domain Incompatibility & OOV Terminology	Scientific, medical, and legal vocabularies diverge sharply from general web corpora (SciBERT shares only 42% vocabulary with base BERT).	Paper 4 shows domain-trained Word2Vec/FastText outperforms general LLMs on domain similarity and classification when context is minimal.

4. Standardized Evaluation Metrics and Protocols

To rigorously assess performance across classification, tokenization, sequence labeling, and semantic embedding tasks, the reviewed papers implement specific mathematical metrics:

- **Exact vs. Relaxed F1-Score:** In sequence labeling (Paper 1), Exact Match requires precise token boundary and entity type alignment ($\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$, $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$, $\text{F1} = 2 * \text{P} * \text{R} / (\text{P} + \text{R})$), while Relaxed Match credits partial spans. Macro-averaging computes metric arithmetic means across classes independently.
- **Non-Parametric Statistical Ranking:** Paper 1 utilizes the Friedman test to evaluate model ranking variance across datasets, followed by Nemenyi post-hoc Critical Difference (CD) analysis to establish statistical significance.
- **Subword Fertility:** Evaluated in Paper 3, fertility measures the average subword tokens generated per natural word (ideal = 1.0). High values (>5.0) indicate severe morphological fragmentation.

- Proportion of Continued Words (PCW) & Normalized Sequence Length (NSL): PCW calculates the fraction of words fragmented into two or more subwords (ideal = 0.0). NSL computes the compression ratio relative to a baseline tokenizer: $NSL = \text{Len}(\text{Tokens_candidate}) / \text{Len}(\text{Tokens_baseline})$.
- Intrinsic Semantic Correlation: Assessed in Paper 4 via Pearson (r), Spearman (rho), and Kendall (tau) correlation coefficients between cosine similarity of dense vectors and human clinical ratings (UMNSRS, SemEval STS).
- Econometric Regression & Contingency Testing: Paper 5 fits Ordinary Least Squares (OLS) multi-variable regression to measure R-squared and aspect coefficient significance against customer star ratings, supported by McNemar chi-squared tests on classification accuracy.

5. Comprehensive Review of the Five Research Papers

5.1 Paper 1: End-to-end Sequence Labeling via Bi-directional LSTM-CNNs-CRF

Authors: Xuezhe Ma, Eduard Hovy

Affiliation: Language Technologies Institute, Carnegie Mellon University, Pittsburgh, PA, USA

Reference / Status: arXiv:1603.01354v5 [cs.CV] (ACL 2016)

Objective & Theoretical Framework: This paper introduces a truly end-to-end neural sequence labeling model that requires no task-specific feature engineering or data preprocessing beyond pre-trained word embeddings. The core architecture combines character-level representation via Convolutional Neural Networks (CNNs), word-level sequential context processing via Bidirectional Long Short-Term Memory (BiLSTM) networks, and sentence-level global label decoding via Conditional Random Fields (CRFs). The framework addresses the limitation of previous models that relied heavily on hand-crafted features or incomplete context representations for sequence labeling tasks like Named Entity Recognition (NER) and Part-of-Speech (POS) tagging.

Methodology & Experimental Design: The proposed architecture features a three-tier neural pipeline:

1. **Character-Level Representation:** A CNN extracts character-level feature vectors from input words to effectively capture morphological information (such as prefixes, suffixes, and capitalization patterns) and mitigate out-of-vocabulary (OOV) issues.
2. **Word-Level Context Encoding:** Character representations are concatenated with pre-trained word embeddings (using Stanford GloVe embeddings for NER and Senna embeddings for POS tagging) and fed into a BiLSTM network to model both left-to-right and right-to-left sequential dependencies.
3. **Global Structured Decoding:** A CRF layer sits atop the BiLSTM outputs to account for global correlations between adjacent output labels (e.g., ensuring valid I-PER transitions only following B-PER or I-PER tags).

The model was evaluated on two benchmark datasets: CoNLL-2003 for Named Entity Recognition (4 entity types: PER, LOC, ORG, MISC) using micro-averaged F1 score, and Penn Treebank (PTB) WSJ for

POS tagging using accuracy. Model parameter training utilized SGD with momentum, dropout regularization applied to both input and output vectors, and learning rate decay.

Key Quantitative Findings:

- **State-of-the-Art Performance on CoNLL-2003 NER:** The model achieved a top-tier F1 score of 91.21% without using external gazetteers or manual feature engineering, outperforming prior feature-heavy baselines (e.g., Chiu & Nichols' 91.62% which used explicit lexicon matching).
- **Superiority on POS Tagging:** On the Penn Treebank WSJ dataset, the BiLSTM-CNNs-CRF architecture attained an accuracy of 97.55%, setting a new state-of-the-art benchmark for English POS tagging at the time of publication.
- **Impact of Integrated Components:** Ablation experiments demonstrated that removing any single component degraded performance significantly:
 - Replacing CRF decoding with a standard softmax classification layer reduced NER F1 by 0.85% (from 91.21% down to 90.36%), proving the necessity of joint sequence decoding.
 - Omitting the character-level CNN module decreased performance by over 1.1% F1, confirming the necessity of sub-word morphological information for unknown word generalization.
- **Robustness Against Unseen Tokens:** The character-level CNN representation allowed the network to accurately tag rare and out-of-vocabulary entities that failed under standard word-embedding lookup baselines.

5.2 Paper 2: Text Classification in the LLM Era — Where Do We Stand?

Authors: Sowmya Vajjala, Shwetal Shimangaud

Affiliation: National Research Council Canada, Ottawa / McGill University, Montreal, Canada

Reference / Status: arXiv:2502.11830v1 [cs.CL] (February 2025)

Objective & Core Questions: This paper investigates the comparative effectiveness of Large Language Models in text classification relative to traditional smaller language models. Evaluating 32 datasets spanning 8 typologically diverse languages across 4 distinct tasks, the study addresses four central research questions: (RQ1) Zero-shot prompting vs. full-data training; (RQ2) Few-shot fine-tuning benefits; (RQ3) Multi-source synthetic data utility; and (RQ4) Supervised instruction fine-tuning vs. classical encoders.

Methodology & Experimental Pipeline: The experimental framework encompasses: (1) Zero-shot prompting of open LLMs (Qwen-2.5-7B, Aya-23-8B, Aya-Expanse-8B) and proprietary GPT-4 (`gpt-4-0613`) using structured Pydantic schemas via the Instructor library; (2) Few-shot fine-tuning with FastFit (10 samples per label using batch contrastive learning with token-similarity scoring) and SetFit; (3) Multi-source synthetic data generation combining GPT-4, Qwen-2.5-7B, and Aya-Expanse-8B in equal balance; and (4) Discriminative classifiers trained on real vs. synthetic data (Logistic Regression over frozen `gte-multilingual-base` embeddings, fine-tuned multilingual BERT, and Instruction Fine-Tuning of Qwen-2.5-7B-Instruct). The 32 datasets cover 8 languages (Arabic, English, French, German, Hindi, Italian,

Portuguese, Spanish) across Twitter Sentiment (3 classes), TAXI1500 (6 biblical topics), Amazon Massive Scenario (18 classes), and Amazon Massive Intent (60 classes).

Key Quantitative Findings:

- Zero-Shot LLMs are Task-Dependent: Zero-shot LLMs perform exceptionally on subjective, low-cardinality tasks like Sentiment (Aya-Expansive-8B: 68% accuracy, GPT-4: 67%), outperforming classifiers trained on full human data (62%). However, on high-cardinality topic tasks (INTENT: 60 classes), zero-shot GPT-4 achieved only 66%, lagging full-data Logistic Regression (82%) and Qwen-2.5 SFT (86%).
- Superiority of Few-Shot FastFit: With only 10 labeled examples per category, FastFit outperformed zero-shot GPT-4 on TAXI1500 (58% vs. 52%) and SCENARIO (72% vs. 63%), executing in just 5–15 minutes on a standard laptop CPU/GPU.
- Multi-Source Synthetic Data Advantage: Classifiers trained on synthetic data generated across multiple LLMs surpassed open zero-shot LLMs on 3 out of 4 tasks and beat zero-shot GPT-4 on SCENARIO (68% vs. 63%), demonstrating a cost-effective, one-time investment alternative.
- Severe Cross-Lingual Disparities: Performance gaps between best- and worst-performing languages averaged 11% to 29% across methods, with Arabic and Hindi suffering severe degradation due to non-Latin script token inflation.

Limitations & Boundary Conditions: The study was restricted to short texts (<50 words); prompting was performed exclusively in English; and all classifiers were monolingual models built on multilingual bases.

5.3 Paper 3: Comparative Analysis of Tokenization Algorithms for Low-Resource Language Dzongkha

Authors: Tandin Wangchuk Hall, Tad Gonsalves

Affiliation: Department of Information and Communication Science, Sophia University, Tokyo, Japan

Reference / Status: arXiv:2509.15255v1 [cs.CL] (September 2025)

Objective & Cultural Motivation: Dzongkha, the national language of Bhutan, is spoken by approximately 700,000 people and is highly resource-constrained. Due to non-segmented Tibetan-derived script characteristics and lack of whitespace word boundaries, pre-trained high-resource tokenizers fail catastrophically. The authors conduct the first rigorous comparative study evaluating subword tokenization algorithms to establish the technical foundation for future Dzongkha LLMs.

Methodology & Experimental Framework: Three candidate subword tokenization algorithms—Byte-Pair Encoding (BPE), WordPiece, and SentencePiece (Unigram)—were trained from scratch on a 27-million character Dzongkha corpus provided by the Dzongkha Development Commission (DDC) across uniform vocabulary sizes of 10,000 and 30,000. Performance was evaluated on a clean benchmark of 180,000 manually segregated Dzongkha words against three state-of-the-art baselines: GPT-2 tokenizer, LLaMA-30B tokenizer, and OpenAI TikToken (o200k base used for GPT-4o). Evaluations measured Normalized Sequence Length (NSL), Subword Fertility, Proportion of Continued Words (PCW), and runtime latency.

Key Quantitative Findings:

- Severe Baseline Fragmentation: Commercial foundation tokenizers exhibited massive over-segmentation on Dzongkha text: GPT-2 had an unusable subword fertility of 13.37; TikToken (GPT-4o) scored 7.19; and LLaMA-30B reached 6.84 tokens per word.

- **SentencePiece Superiority:** Dedicated SentencePiece (Unigram) achieved the lowest subword fertility of 0.79 (ideal is 1.0, reflecting multi-word subword compression) and the lowest Proportion of Continued Words (PCW) of 0.09 (breaking only 15,821 out of 180,000 words), compared to BPE (PCW: 0.13, fertility: 0.93) and WordPiece (PCW: 0.27, fertility: 1.35).
- **Sequence Compression Efficiency:** SentencePiece yielded the lowest NSL values across all baselines: 0.0594 relative to GPT-2, 0.1162 relative to LLaMA-3, and 0.1105 relative to TikToken, compressing sequences by nearly 90%.
- **Inference Speed:** SentencePiece executed encoding in 131 ms per loop on the 180,000-word dataset, outperforming WordPiece (386 ms), BPE (439 ms), GPT-2 (218 ms), and LLaMA-30B (690 ms).

Limitations & Future Work: SentencePiece exhibited slower vocabulary initialization (approx. 2 min vs. 2 s for WordPiece); downstream validation on machine translation and classification remains an open milestone.

5.4 Paper 4: Evaluating Embedding Frameworks for Scientific Domain

Authors: Nouman Ahmed, Ronin Wu, Victor Botev

Affiliation: Iris.ai / University of Oxford / QunaSys

Reference / Status: arXiv:2510.06244v1 [cs.CL] (October 2025)

Objective & Scientific Scope: Developing optimal word representations for scientific literature is uniquely challenging due to dense technical terminology and out-of-vocabulary (OOV) tokens. While domain Transformers like SciBERT yield contextual embeddings, training them requires extensive compute (e.g., TPU 8-core cluster for one week). This paper builds a multi-task evaluation suite to assess whether lightweight, static embedding models (Word2Vec, FastText) trained on domain corpora can rival or outperform heavy Transformer models.

Methodology & Multi-Task Suite: The authors constructed an evaluation suite covering both intrinsic and extrinsic benchmarks: (1) Word Similarity: UMNSRS (566 medical concept pairs with 189 OOV pairs); (2) Sentence Similarity: SemEval-2017 (general), Clinical STS 2018 (1,068 pairs), and Clinical STS 2019 (2,054 pairs); (3) Named Entity Recognition: CoNLL-2003 (general), CHEMDNER (84,355 chemical entities), and SciERC (8,098 AI entities) evaluated via BiLSTM sequence decoders; and (4) Document Classification: Reuters-12 (general) and BioChem-8 (39,845 biomedical papers across 8 classes) evaluated via k-NN (k=3). Models evaluated included Word2Vec and FastText (CBOW and Skipgram, 100D and 200D) trained on 2.5 million scientific abstracts from CORE, compared against BERT, RoBERTa, GPT-2, and SciBERT.

Key Quantitative Findings:

- **Word & Sentence Similarity:** Word2Vec Skipgram (200D) trained on domain abstracts dominated sentence similarity across all benchmarks, scoring Pearson $r = 0.6749$ on SemEval, 0.7588 on Clinical STS 2018, and 0.6362 on Clinical STS 2019, consistently outperforming BERT (0.5262) and SciBERT (0.4705 on SemEval, 0.6752 on STS 2018).
- **Document Classification:** Word2Vec Skipgram (200D) achieved the top F-beta on Reuters-12 (0.9243) and BioChem-8 (0.9015), exceeding SciBERT (0.8768 and 0.8423) and RoBERTa (0.8338 and 0.8085).
- **Extrinsic NER Exception:** On token-level NER, Transformer architectures outperformed static

embeddings due to bidirectional contextualization and subword handling. SciBERT led on CHEMDNER (0.5399 F-beta) and SciERC (0.3841 F-beta), while RoBERTa led CoNLL-2003 (0.8307 F-beta).

- Subword Tokenization on OOV: On UMNSRS, standard Word2Vec returned NaN for 189 OOV pairs. FastText Skipgram with character n-grams achieved the most robust overall Pearson score (0.4737), seamlessly bridging in-vocabulary and OOV representations.

Limitations & Practical Implications: The authors recommend a hybrid strategy: leveraging lightweight whole-word Word2Vec embeddings for similarity and document retrieval, backed by subword FastText/WordPiece representations for rare terminology and token-level entity extraction.

5.5 Paper 5: Beyond the Star Rating: A Scalable Framework for Aspect-Based Sentiment Analysis Using LLMs and Text Classification

Authors: Vishal Patil, Shree Vaishnavi Bacha, Revanth Yamani, Yidan Sun, Mayank Kejriwal

Affiliation: University of Southern California (USC), Information Sciences Institute, Marina del Rey, CA, USA

Reference / Status: arXiv:2602.21082v1 [cs.CL] (February 2026)

Objective & Industrial Scalability: Online customer reviews contain granular aspect feedback (food, service, ambiance) that overall star ratings obscure. While LLMs can extract nuanced aspect sentiments, prompting commercial models over millions of reviews is cost-prohibitive. The authors design a scalable neuro-symbolic framework coupling LLM zero-shot aspect discovery with fast, off-the-shelf classical machine learning, deploying the pipeline across 4.7 million Yelp reviews collected over 17 years.

Methodology & Econometric Modeling: The workflow consists of: (1) Aspect Discovery: Querying ChatGPT-4o and ChatGPT-4o-mini on 600 sampled reviews across 30 restaurants to isolate six recurring aspects (Service, Food Quality, Ambiance, Wait Time, Price, Menu Variety), validated against LDA topic modeling; (2) Manual Annotation & Agreement: Human labeling of 5,000 reviews across the 6 aspects (-1 negative, 0 neutral/irrelevant, +1 positive), achieving substantial Fleiss' Kappa (Price: 0.943, Food: 0.774, Service: 0.727); (3) Scalable Classifier Comparison: Evaluating Multinomial Naive Bayes, Logistic Regression, Random Forest, and SVM over TF-IDF and 100D fastText embeddings; (4) Architecture Design: Benchmarking a One-Stage multi-class model vs. a Two-Stage (Relevance -> Sentiment) model, with one-sided McNemar tests confirming One-Stage superiority; and (5) Multi-variable OLS Regression: Modeling overall restaurant rating as a function of the 6 predicted aspect sentiments, controlling for 10 cuisine types and 18 regional states.

Key Quantitative Findings:

- Model Selection: Logistic Regression with fastText embeddings achieved 76.6% average accuracy across all six aspects, providing superior computational throughput over SVM (which was intractable on 4.7M reviews).
- Explanatory Power of Extracted Aspects: Across 52,286 restaurants, the regression models demonstrated remarkable goodness-of-fit ($R^2 = 0.814$ to 0.819), proving that machine-labeled aspect sentiments explain over 81% of restaurant rating variance.
- Aspect Importance Hierarchy: Food Quality was the strongest rating determinant ($\beta = 1.58$ to 1.59 , $p < 0.001$), followed by Service ($\beta = 0.74$ to 0.78 , $p < 0.001$) and Menu Variety ($\beta = 0.66$ to 0.70 , p

< 0.001). Price was statistically non-significant ($p > 0.05$).

- Counter-Intuitive Behavioral Insights: Wait Time exhibited a statistically significant positive coefficient (beta = 0.18 to 0.24, $p < 0.001$), supporting 'social proof' theory where long wait times signal high popularity and perceived dining value. Ambiance showed a negative association (beta = -0.27 to -0.31, $p < 0.001$), reflecting elevated customer expectations in upscale settings.

6. Cross-Paper Comparative Analysis & Synthesis

The five surveyed papers represent complementary pillars of the modern NLP pipeline: tokenization and vocabulary engineering (Paper 3), semantic embedding representation (Paper 4), structured sequence labeling (Paper 1), multi-class text classification (Paper 2), and scalable industrial sentiment extraction (Paper 5). Tables 2, 3, and 4 synthesize their core dimensions, architectures, and empirical findings.

Table 2: Cross-Study Architectural and Methodological Matrix

Paper Reference	Core NLP Task	Primary Models Evaluated	Vectorization / Tokenization	Dataset Scale & Domains
Paper 1 (Keraghel et al., 2024)	Named Entity Recognition (NER)	CRF, LSTM-CRF, BERT, RoBERTa, DeBERTa-v3, GLiNER, GPT-4o	BPE, WordPiece, RoBERTa tokenizer, BIO/IOBES tags	10 datasets (CoNLL, OntoNotes, Few-NERD, BioNLP, FIN, etc.)
Paper 2 (Vajjala & Shimangaud, 2025)	Multi-Class Text Classification	GPT-4, Qwen-2.5-7B, Aya-23, Aya-Expanse, FastFit, mBERT, SFT	BPE, TikToken, gte-multilingual-base (305M)	32 datasets across 8 languages (Sentiment, Bible, Voice Intents)
Paper 3 (Hall & Gonsalves, 2025)	Subword Tokenization Evaluation	SentencePiece (Unigram), Byte-Pair Encoding (BPE), WordPiece	Subword tokenizers (10k, 30k vocab) vs. TikToken, LLaMA-3, GPT-2	27M char corpus; 180k Dzongkha words benchmark
Paper 4 (Ahmed et al., 2025)	Domain Embedding Evaluation	Word2Vec, FastText, BERT, RoBERTa, GPT-2, SciBERT	CBOW/Skipgram static vectors (100D/200D), BPE, WordPiece	2.5M CORE abstracts; 9 intrinsic/extrinsic NLP benchmarks
Paper 5 (Patil et al., 2026)	Aspect-Based Sentiment & Regression	ChatGPT-4o/mini, Logistic Regression, SVM, Random Forest, OLS	TF-IDF (400D), FastText custom embeddings (100D)	4.72M Yelp reviews, 52,286 restaurants, 17 years of data

Table 3: Primary Empirical Findings Across Surveyed Studies

Paper Reference	Key Evaluation Metrics	Leading Model / Configuration	Critical Empirical Insight
Paper 1 (NER Survey)	Exact Macro F1, Precision, Recall, Friedman CD	DeBERTa-v3-large (91.36 F1 general); CRF (90.94 F1 disease)	Zero-shot GPT-4o (65.17 F1) severely lags task-fine-tuned encoders; CRFs excel in low-data biomedicine.
Paper 2 (Text Classification)	Accuracy, Cross-lingual Disparity, Training Time	FastFit (few-shot topic); Qwen SFT (full-data intent)	Zero-shot LLMs win only on 3-class sentiment; few-shot FastFit beats GPT-4 on topic tasks with 10 samples.
Paper 3 (Dzongkha Tokenization)	Subword Fertility, PCW, NSL, Execution Latency	SentencePiece Unigram (Fertility: 0.79, PCW: 0.09, 131 ms)	Commercial LLM tokenizers fragment Dzongkha severely (GPT-2: 13.37 fertility; GPT-4o: 7.19 fertility).
Paper 4 (Scientific Embeddings)	Pearson r, Spearman rho, Downstream F-beta	Word2Vec 200D Skipgram (similarity/classification); SciBERT (NER)	Static domain embeddings outperform Transformers on word/sentence similarity and topic classification.
Paper 5 (ABSA & Rating Regression)	Accuracy, F1-Score, Regression R^2, Effect Betas	FastText + Logistic Regression (76.6% Acc; R^2: 0.819)	Machine aspect sentiments explain 81.9% of restaurant star ratings; food & service dominate; wait time is positive.

7. Critical Research Gaps and Future Directions

Synthesizing the limitations across all five papers reveals several unaddressed frontiers in NLP research:

1. Morphological Tokenization in Indigenous Languages: Paper 3 highlights that high-resource tokenizers fail on Dzongkha, but evaluation was limited to intrinsic segmentation. Future research must measure downstream impacts when training native Dzongkha encoders for machine translation, scam detection, and factual question answering.
2. The High-Cardinality Multi-Label Frontier: Paper 2 demonstrated the collapse of zero-shot LLMs on 60-class single-label classification. Extending this benchmarking to extreme multi-label text classification (XMLC, with thousands of sparse categories) is urgently needed.
3. Bridging Domain Static Vectors with Modern Contextual Encoders: Paper 4 revealed that static Word2Vec outperformed Transformers on similarity benchmarks. Developing hybrid distillation

techniques that transfer domain vocabulary coverage into compact contextual encoders without requiring weeks of TPU pre-training remains an open challenge.

4. Dynamic, Multi-Modal ABSA at Industrial Scale: Paper 5 extracted six static dining aspects from Yelp text. Real-world platforms require dynamic aspect discovery capable of tracking evolving consumer expectations across multi-modal assets (text reviews, user images, and dish menus).

8. Conclusion and Strategic Architectural Recommendations

The comprehensive synthesis of these five foundational studies confirms that the paradigm of prompting monolithic LLMs for all NLP tasks is sub-optimal across accuracy, computational efficiency, and economic cost. The findings support a set of concrete architectural principles for NLP engineers:

- For Sequence Labeling (NER): Prioritize dedicated bidirectional Transformer architectures (DeBERTa-v3, GLiNER) or classical CRF heads for specialized vocabularies over generative LLM prompting.
- For High-Throughput Classification: Use zero-shot LLMs only for subjective, low-cardinality tasks. In topic classification with >5 classes, collect 10 examples per label and deploy contrastive few-shot models (FastFit) or fine-tune task encoders on multi-source synthetic data.
- For Low-Resource & Non-Latin Languages: Build dedicated SentencePiece (Unigram) subword vocabularies from raw native corpora to prevent catastrophic token fragmentation and inflated inference costs.
- For Specialized Scientific Retrieval: Leverage domain-adapted static embeddings (Word2Vec/FastText) for similarity clustering and document ranking, reserving Transformers for token-level information extraction.
- For Massive Industrial ABSA: Implement neuro-symbolic hybrid workflows—harnessing LLMs offline for aspect taxonomy discovery, while training lightweight discriminative classifiers (fastText + Logistic Regression) for production inference across millions of documents.

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